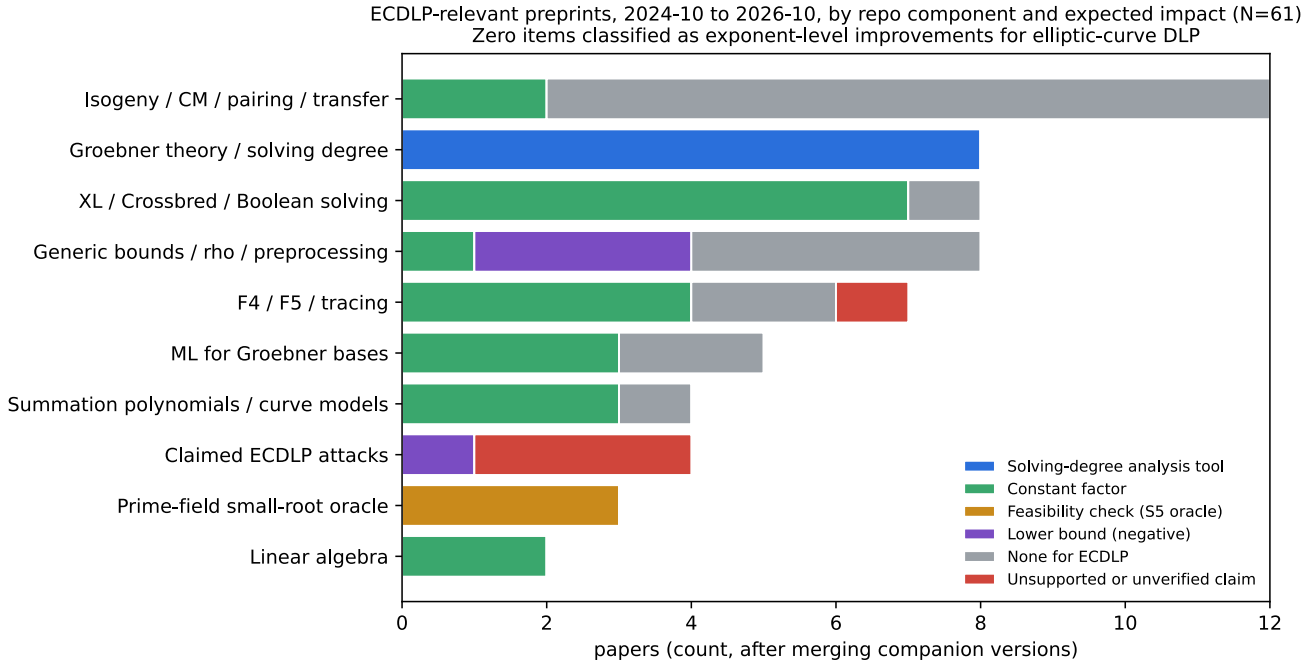
1. Literature search, 2025–26: ECDLP, index calculus, Gröbner solving

Date 2026-10-11 · **Type** literature search round, no experiment run · **Status** every statement is *proposed*, taken from abstracts that were opened and checked; nothing measured.

Headline (negative). None of the 61 classified ePrint/arXiv papers (2024-10 to 2026-10) claims or plausibly delivers an exponent-level improvement for ECDLP. Nothing in the window addresses summation polynomials, point decomposition, Weil descent for ECDLP, or GHS. No new break or trapdoor was found for P-curves, secp256k1, Brainpool or the sect curves.



Source: research/literature_search_20261011/papers.csv; classification from abstracts (search round 2026-10-11).

Figure 1: Papers by repo component and expected impact, N = 61, counts from `papers.csv`. The classification is a judgement made from the abstracts.

1.1. What matters most

- **Structured generic-group model** (ePrint 2026/384, EUROCRYPT 2026). Algorithms that exploit structure on a δ fraction of elements need time Ω(min(√q, 1/δ)), with analogous bounds stated for EC points. This formalises the repo's "Koblitz IC is generic" verdict.
- **Non-adaptive preprocessing bound** (arXiv 2505.00894, STOC 2026). Non-adaptive advice cannot beat √N online time unless S = Ω(√N).
- **Solving-degree diagnostics** for descended S_{m+1}:
  - Kudo–Yokoyama, saturation exponent and first-fall synchronisation (ePrint 2024/528, 2025/822).
  - Multi-homogeneous XL (ePrint 2025/2060).
  - Symmetric-function solving-degree proofs (arXiv 2609.14052).
  - Generic leading monomials (arXiv 2505.10246).
- **Constant-factor engineering:**
  - lane-batched tracing (arXiv 2607.06372);
  - early abort on inconsistency (arXiv 2603.24890);
  - Crossbred trimming (ePrint 2026/360, 2026/408, 2024/758);
  - Wiedemann-XL (ePrint 2026/1787);
  - TNFS Galois k^2 linear algebra (ePrint 2026/560), likely already captured by GGMP n^2
  - binary-curve x-only isogenies (ePrint 2026/704);
  - co-Z table-free BSGS (ePrint 2026/2102).
- **S5 small-root oracle feasibility on paper:** cuso (ePrint 2024/1577) and the LP asymptotics (ePrint 2026/1027).
- **Flagged as unsupported:** ePrint 2025/015, arXiv 2607.09814, a 16-bit D-Wave ECDLP result, arXiv 2607.09163.

1.2. Proposed falsifiers (ranked)

1. Saturation exponent and first degree fall against the measured solving degree, descended S3/S4, n = 11–23.
2. Multi-graded Hilbert function of descended S3 with point blocks, n ≈ 15.
3. δ table: measured IC cost against min(√r, 1/δ) for every configuration. A cost below the bound would indicate an accounting bug.
4. Early abort at the F4 refutation degree.

5. Lane-batched inherited-F4 replay, k = 4–8.
6. E/G check: FGHR-symmetrised S_m on E against x-only S_m on $E/\langle T \rangle$.
7. LP/cuso asymptotic bound for S5. A bound below $p^{1/5}$ closes the route.

Canonical graphs checked and unchanged, because there is no new measured value: the IC scoreboard, `progress-timeline.json`, `docs/curves/`, `docs/performance-gains/` and `figures/`. Coverage caveats: ePrint search matches titles and keywords and was rate-limited (HTTP 429); ePrint 2026/1198 and 2025/271 were not opened. The full discussion is in `README.md`.

1.3. Classified papers

ID	Paper	Date	Component	Impact
ePrint 2026/384	Corrigan-Gibbs, Henzinger, Wu: The Structured Generic-Group Model	2026-02-24	Generic bounds / rho / preprocessing	Lower bound (negative)
ePrint 2025/783 = arXiv 2505.00894	Dinur, Keller, Marmor: Non-adaptive time-space lower bounds (Shearer-like inequality)	2026-05-20	Generic bounds / rho / preprocessing	Lower bound (negative)
ePrint 2026/2253	Akshima et al.: Generic bounds for multi-instance problems	2026-09-28	Generic bounds / rho / preprocessing	Lower bound (negative)
ePrint 2024/564	May, Ostuzzi: Multiple group action dlogs with(out) precomputation	2025-10-03	Isogeny / CM / pairing / transfer	None for ECDLP
ePrint 2024/528	Kudo, Yokoyama: Degree bounds and synchronization for affine semi-regular systems	2026-09-02	Groebner theory / solving degree	Solving-degree analysis tool
arXiv 2410.23211 = ePrint 2025/822	Kudo, Yokoyama: Generalization of semi-regular sequences	2025-05-08	Groebner theory / solving degree	Solving-degree analysis tool
ePrint 2024/1924	Gaggero, Gorla: The complexity of solving a random polynomial system	2024-11-27	Groebner theory / solving degree	Solving-degree analysis tool
arXiv 2609.14052 = ePrint 2026/1995	Gorla, Trebiani: Symmetric models for syndrome decoding	2026-09-12	Groebner theory / solving degree	Solving-degree analysis tool
ePrint 2025/2060	Ning, Ran, Samardjiska: Multi-homogeneous XL	2025-11-07	Groebner theory / solving degree	Solving-degree analysis tool
ePrint 2025/793	Nakamura: Solving systems via Macaulay matrices	2026-03-06	Groebner theory / solving degree	Solving-degree analysis tool
arXiv 2505.10246	Sakata, Takagi: Leading monomials of a minimal GB of generic sequences	2025-05-15	Groebner theory / solving degree	Solving-degree analysis tool
arXiv 2502.12721	Bardet, Gilard: Hilbert series of Support-Minors (with ePrint 2025/1083; arXiv 2609.07223)	2025-02-18	Groebner theory / solving degree	Solving-degree analysis tool
arXiv 2609.18866 = ePrint 2026/2062	La Scala, Marchesin, Tiwari: Hamming ideals and GB for ISD-like decoding	2026-09-16	Summation polynomials / curve models	Constant factor
ePrint 2026/1299	Drylo: Decomposition of compressions on elliptic curves	2026-06-22	Summation polynomials / curve models	Constant factor
arXiv 2608.01675	Wu: A characteristic-uniform model for elliptic curves	2026-08-17	Summation polynomials / curve models	None for ECDLP
arXiv 2603.16378	Kouba, Neiger, Safey El Din: F4 with tracer data complexity	2026-03-17	F4 / F5 / tracing	Constant factor
arXiv 2609.14433	Demin: Probably correct row echelon form in F4	2026-09-13	F4 / F5 / tracing	Constant factor
arXiv 2607.06372	Demin: Groebner.jl fast tracing (SIMD-packed coefficients)	2026-08-01	F4 / F5 / tracing	Constant factor
arXiv 2601.05026	Lairez, Mohr, Ternier: Monomial-ideal data structure for signature GB	2026-01-08	F4 / F5 / tracing	Constant factor
arXiv 2607.09163	Ma, Liu, Jiao: Proper-Cover algorithm beyond F5/GVW	2026-07-10	F4 / F5 / tracing	Unsupported or unverified claim
arXiv 2601.06765	Gokavarapu: GPU reductions for multivariate systems (design only)	2026-01-11	F4 / F5 / tracing	None for ECDLP
ePrint 2026/2364	Holy, Jurecek: Open-source GB implementations on small-scale AES over GF(2)	2026-10-05	F4 / F5 / tracing	None for ECDLP
ePrint 2026/408	Jaques, Ran, Samardjiska, Seitner: Smoothing the degree of regularity	2026-10-07	XL / Crossbred / Boolean solving	Constant factor
ePrint 2024/758	Baena et al.: Admissible parameters for Crossbred	2025-05-23	XL / Crossbred / Boolean solving	Constant factor
ePrint 2026/360	Vidal, Delaplace, Ionica: Improved preprocessing for Crossbred	2026-02-22	XL / Crossbred / Boolean solving	Constant factor
ePrint 2026/1704	Bouillaguet, Sauvage: Decrementally-improved Boolean MQ	2026-08-16	XL / Crossbred / Boolean solving	Constant factor
ePrint 2026/040	Liu et al.: Evaluation on bounded-Hamming-weight inputs (with ePrint 2026/197)	2026-09-16	XL / Crossbred / Boolean solving	Constant factor
arXiv 2506.09950 = ePrint 2025/1097	La Scala, Tiwari: Oracle-based multistep solving	2025-06-11	XL / Crossbred / Boolean solving	Constant factor

ID	Paper	Date	Component	Impact
ePrint 2025/1788	May, Ostuzzi, Ressler: Just Guess (underdetermined MQ; with ePrint 2026/1054)	2025-09-30	XL / Crossbred / Boolean solving	None for ECDLP
arXiv 2603.24890	Horak, Semaev: Inconsistency probability of sparse equations over F2	2026-03-26	XL / Crossbred / Boolean solving	Constant factor
ePrint 2026/1787	Chou, Niederhagen: Practical optimization for Wiedemann XL (with ePrint 2026/1382)	2026-08-24	Linear algebra	Constant factor
ePrint 2026/560	Al Aswad, Pierrot, Thome: High-order Galois automorphisms for TNFS linear algebra	2026-06-19	Linear algebra	Constant factor
ePrint 2026/1927	Bariant: Forgery attack against Frobenius-UOV (linearized resultant)	2026-09-08	Summation polynomials / curve models	Constant factor
ePrint 2024/1577	Ryan: Multivariate Coppersmith with known moduli (cuso)	2025-03-01	Prime-field small-root oracle	Feasibility check (S5 oracle)
ePrint 2026/1027	Ding et al.: Asymptotic bounds for automated Coppersmith via LP	2026-09-18	Prime-field small-root oracle	Feasibility check (S5 oracle)
ePrint 2026/423	Ding et al.: Coppersmith for MIHNP via determinant-based elimination	2026-03-02	Prime-field small-root oracle	Feasibility check (S5 oracle)
ePrint 2025/1243	Galbraith, Gilchrist, Robert: Ascending isogeny volcanoes	2025-09-20	Isogeny / CM / pairing / transfer	None for ECDLP
ePrint 2024/1556	Robert: The module action for isogeny based cryptography	2026-03-16	Isogeny / CM / pairing / transfer	None for ECDLP
ePrint 2026/704	Banegas, Sarkis, Smith: Fast isogeny evaluation on binary curves	2026-05-21	Isogeny / CM / pairing / transfer	Constant factor
ePrint 2025/1652	Zheng, Lin, Zhao: Pairings with embedding degree two via biextensions	2025-09-12	Isogeny / CM / pairing / transfer	Constant factor
ePrint 2025/1706	Moriya: Kani's lemma from Clifford algebra	2025-09-19	Isogeny / CM / pairing / transfer	None for ECDLP
ePrint 2025/1514	de Boer, Pellet-Mary, Wesolowski: Rigorous methods for computational number theory	2026-02-19	Isogeny / CM / pairing / transfer	None for ECDLP
arXiv 2604.11330	Ghitza, Gupta, Kortge: Inverse volcano problem over non-prime fields	2026-05-28	Isogeny / CM / pairing / transfer	None for ECDLP
arXiv 2605.07626	el Baraka, Ezzouak: Weighted distributions of CM orders	2026-05-08	Isogeny / CM / pairing / transfer	None for ECDLP
arXiv 2503.15428	Stange: Division polynomials for arbitrary isogenies	2026-04-17	Isogeny / CM / pairing / transfer	None for ECDLP
arXiv 2512.09601	Au-Yeung: Valuation of elliptic nets for CM curves	2025-12-10	Isogeny / CM / pairing / transfer	None for ECDLP
ePrint 2026/110	Barbulescu et al.: Density of rank>=1 genus-2 Jacobians	2026-01-23	Isogeny / CM / pairing / transfer	None for ECDLP
ePrint 2025/1079	Bollauf, Parisella, Siim: Revisiting discrete logarithm reductions	2025-10-07	Generic bounds / rho / preprocessing	None for ECDLP
ePrint 2026/1823	Panny et al.: HyperSolver (Delfs-Galbraith, isogeny ladders)	2026-08-27	Generic bounds / rho / preprocessing	None for ECDLP
ePrint 2026/2102	Cenk et al.: Windowed-batch-inversion small ECDLP solver	2026-09-18	Generic bounds / rho / preprocessing	Constant factor
arXiv 2508.14011	Dallaire-Demers, Doyle, Foo: ECDLP challenge ladder	2026-03-25	Generic bounds / rho / preprocessing	None for ECDLP
arXiv 2505.23696	Kera et al.: Oracle border basis (transformer oracles)	2025-05-29	ML for Groebner bases	Constant factor
arXiv 2512.14722 = ePrint 2025/2219	Malhou, Perret, Lauter: HATSolver	2025-12-09	ML for Groebner bases	None for ECDLP
arXiv 2602.02972	Bunch et al.: Learning fast monomial orders	2026-02-03	ML for Groebner bases	Constant factor
arXiv 2606.07321	Shafiq et al.: Homogeneity entropy S-pair selection	2026-06-05	ML for Groebner bases	None for ECDLP
arXiv 2509.14754	Luo, Sun, Long: Variable ordering for Boolean characteristic sets	2025-09-18	ML for Groebner bases	Constant factor
ePrint 2025/015	Hu: A new method for discrete logarithm based on index calculus	2025-09-02	Claimed ECDLP attacks	Unsupported or unverified claim
ePrint 2024/1923	Hu: Index calculus on prime-field elliptic curves (concludes no subexponential method)	2024-11-27	Claimed ECDLP attacks	Lower bound (negative)
arXiv 2607.09814	Mahalanobis: Guess-and-determine attack on ECDLP (zero minors)	2026-08-06	Claimed ECDLP attacks	Unsupported or unverified claim
DOI 10.23919/ICN.2026.0017	Lin et al.: ECDLP via quantum tunnelling on D-Wave (16-bit)	2026-09-21	Claimed ECDLP attacks	Unsupported or unverified claim
arXiv 2506.12844	Rudolph: Iteration maps for parallel Pollard rho (integer factoring)	2025-06-15	Generic bounds / rho / preprocessing	None for ECDLP